

Microprocessors and Microcontrollers

Assignment 10- Week 10

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15 M

QUESTION 1:

Correct Answer: c

Detailed Solution: The ARM THUMB instruction set is designed to provide higher code density by using 16-bit instructions instead of the standard 32-bit ARM instructions. This allows for reduced memory usage while maintaining reasonable performance. However, in THUMB-2, both 16-bit and 32-bit instructions are used for better flexibility and efficiency.

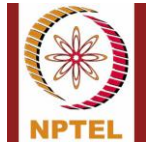
QUESTION 2:

Correct Answer: b

Detailed Solution:

PIC18F microcontrollers use the Harvard architecture, which separates the memory for instructions and data. This allows simultaneous access to both instruction and data memory, improving performance compared to the Von Neumann architecture, which uses a shared memory space for both. The Harvard architecture is commonly used in microcontrollers for faster execution and efficiency.

- Von Neumann architecture (option a) uses a single memory for both instructions and data, leading to a bottleneck.
 - Gordon Moore (option c) is known for Moore's Law, which predicts the exponential growth of transistors on a chip.
 - Alan Turing (option d) is a pioneer in computer science, known for the Turing machine concept, but not related to microcontroller architecture.
-



QUESTION 3:

Correct Answer: a

Detailed Solution: The PIC18F series microcontrollers have a 21-bit address bus for accessing program memory. This allows them to address up to 2MB (2^{21} locations) of program memory. The large address space enables the microcontroller to handle more complex programs and instructions efficiently.

- 12-bit (option b) and 8-bit (option d) address buses are too small to address the memory required for PIC18F microcontrollers.
- 16-bit (option c) address bus is used in some microcontrollers but is not sufficient for PIC18F's full memory addressing capabilities.

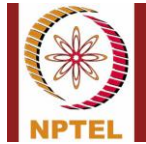
QUESTION 4:

Correct Answer: c

Detailed Solution: The PIC18F family of microcontrollers follows a Harvard architecture, which means it has separate buses for program memory (DPM) and data memory (DDM).

- The program memory bus (DPM) is 16-bit wide, allowing it to fetch a complete instruction in a single cycle.
- The data memory bus (DDM) is 8-bit wide, as the CPU operates on 8-bit data at a time.

This architecture enhances performance by allowing simultaneous instruction fetching and data handling.



QUESTION 5:

Correct Answer: c

Detailed Solution: The MOVLW (Move Literal to W) instruction in PIC18F microcontrollers is used to load an 8-bit literal value (constant) into the WREG register.

The instruction format for MOVLW k (where k is an 8-bit literal) is:

0000 1110 kkkk kkkk

- The opcode for MOVLW is 0000 1110.
- The literal value 0x08 (which is 0000 1000 in binary) is placed in the lower 8 bits.

Thus, the binary encoding of MOVLW 0x08 is:
0000 1110 0000 1000, which corresponds to option (c).

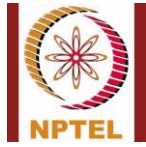
QUESTION 6:

Correct Answer: a

Detailed Solution:

The AVR AT90S2313 is an 8-bit microcontroller from Atmel's AVR family, and its specifications include:

- 2K bytes of In-System Programmable (ISP) Flash memory
 - 128 bytes of SRAM
 - 128 bytes EEPROM
 - 15 I/O lines
 - UART, timers, and watchdog support
-



QUESTION 7:

Correct Answer: b

Detailed Solution: AVR microcontrollers were developed by Alf Egil Bogen and Vegard Wollan, two Norwegian engineers from Atmel Corporation in the 1990s. AVR is a family of RISC-based microcontrollers widely used in embedded systems and known for their efficiency and ease of use, especially in platforms like Arduino.

QUESTION 8:

Correct Answer: c

Detailed Solution: The AT90S2313 microcontroller, like other AVR microcontrollers, follows the RISC (Reduced Instruction Set Computing) architecture and includes 32 general-purpose registers (R0–R31).

These registers:

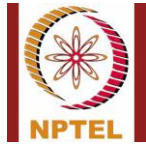
- Are directly connected to the ALU (Arithmetic Logic Unit) for fast processing.
- Support single-cycle access for high-speed operations.
- Provide efficient instruction execution, minimizing memory accesses.

QUESTION 9:

Correct Answer: d

Detailed Solution:

In AVR microcontrollers, the ADD Rd, Rr instruction has the following 16-bit opcode format:



0000 11rd dddd rrrr

Where:

- dddd represents the destination register (Rd).
- rrrr represents the source register (Rr).
- rd are additional bits used to determine register pairs.

Now, breaking down 0x0CC8:

1. Hex to Binary Conversion:

0x0CC8 = 0000 1100 1100 1000

1.
 - The first 6 bits (000011) confirm that this is an ADD instruction.
 - The next 5 bits (01100) represent Rd = R6.
 - The last 5 bits (01000) represent Rr = R8.

Thus, the instruction corresponding to 0x0CC8 is:

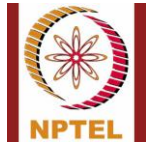
ADD R6, R8,

QUESTION 10:

Correct Answer: d

Detailed Solution: Register Direct Addressing refers to instructions that operate directly on registers (R0–R31) without involving memory addresses.

- Option (a) - EOR R23:
 - EOR (Exclusive OR) is a register-direct instruction.
 - It operates on two registers (implicitly: EOR Rd, Rr), meaning EOR R23, R23 is valid.
- Option (b) - ADD R0, R10:
 - The ADD instruction is register-direct because it performs $R0 = R0 + R10$.
- Option (c) - CLR R22:
 - The CLR (Clear Register) instruction is actually EOR R22, R22, which is register-direct.
- Option (d) - ST Z, R14 (Incorrect for Register Direct Addressing):
 - ST (Store) is an indirect addressing instruction.



- It stores the value from R14 to the memory location pointed by the Z register (R30:R31).
- Since it involves memory access, it does not follow register direct addressing.

QUESTION 11:

Correct Answer: b

Detailed Solution:

Timer 0 (8-bit Timer):

- The Output Compare Register OCR0 is 8 bits wide, as Timer 0 is an 8-bit timer in AVR microcontrollers.

Timer 1 (16-bit Timer):

- The Output Compare Register OCR1 is 16 bits wide, as Timer 1 is a 16-bit timer in AVR microcontrollers.

QUESTION 12:

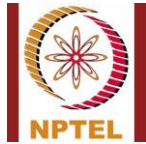
Correct Answer: d

Explanation:

Statement 1: "If memory is organized as 16-bit words, ARM code is 25% faster than THUMB."

Incorrect. ARM uses 32-bit instructions, which are not optimized for 16-bit memory, making ARM slower in this case.

Statement 2: "If memory is organized as 32-bit words, ARM code is 45% faster than THUMB."



Incorrect. While ARM benefits from 32-bit memory alignment, the speedup is not necessarily 45%, and THUMB code is designed for smaller memory footprints.

Statement 3: "If memory is organized as 16-bit words, THUMB code is 45% faster than ARM."

Correct. Since THUMB uses 16-bit instructions, it performs much better than ARM in a 16-bit memory system, where ARM would need extra cycles for instruction fetching.

Statement 4: "If memory is organized as 32-bit words, THUMB code is 60% faster than ARM."

Incorrect. ARM benefits from a 32-bit memory system, and THUMB would not be significantly faster in this case.

QUESTION 13:

Correct Answer: b

Detailed Solution:

ARM Instruction Set: The ARM architecture is designed for 32-bit instructions, providing higher performance than THUMB, which uses 16-bit instructions.

32-bit Memory: A 32-bit memory system allows fetching full ARM instructions in a single cycle, improving speed and efficiency.

THUMB Code: While it improves code density (reduces memory usage), it generally does not offer the best performance compared to ARM instructions.

QUESTION 14:

Correct Answer: c

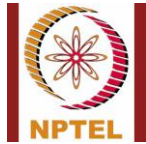
Detailed Solution:

In ARM processors, the SWI (Software Interrupt) instruction is used for invoking system calls.

In ARM mode, the SWI instruction allows a 24-bit immediate operand, enabling 16.7 million different calls.

In THUMB mode, the SWI instruction is only 16 bits long.

The opcode structure in THUMB mode allows for an 8-bit immediate operand (0-255). This means a maximum of 256 different SWI calls can be made in THUMB mode.



QUESTION 15:

Correct Answer: b

Detailed Solution:

The smallest microcontroller in the AVR family refers to the one with the fewest features and smallest memory size.

- ATMEGA128 is a high-end AVR microcontroller with 128KB Flash, 4KB EEPROM, and many peripherals.
- ATTiny24 and ATTiny22 are small AVR microcontrollers but still have more features than ATTiny11.
- ATTiny11 is the smallest in terms of memory and functionality. It has:
 - 1 KB Flash Memory
 - 32 Bytes RAM
 - No EEPROM
 - Limited peripherals (no ADC, no interrupts)

Thus, ATTiny11 is the smallest microcontroller in the AVR family.

*****END*****